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Group 35

Tasks & Contributions

I contributed mainly to the reporting/documentation part, and the final verification and debugging phase of the project.

I also prepared and improved the written deliverables, especially the sections explaining the architecture, the design choices and how each layer works. I also helped to translate our technical implementation into a clean story that meets the project requirements including modules, Result objects, persistence, output files and statistics format.

Collaboration & Challenges

We worked closely together, dividing the work. I made sure the system behavior, outputs and report were in line with the requirements, while my teammate took care of implementation heavy chores. The hardest part was finding the subtle problems that only show up after running things a few times: persistence behavior, file cleanup rules, making sure the “failure conditions” are properly recorded without crashing.

Self-Assessment

This project gave me a better understanding of DBMS internals beyond SQL, especially on importance of modularity and division of duties across layers. I also improved my debugging skills for systems code by developing minimum replicable tests, checking accuracy across different conditions, and verifying outputs against specifications.

Use AI Tools

I mostly used AI technologies for help with writing and learning:

- Clarified Concepts: I asked questions to better understand DBMS concepts, such as buffer management, page I/O accounting, B+-tree behavior, and how to interpret the experiments.

- Report Wording & Structuring: Helped to reword explanations and structure sections of the report to the required format with the use of AI.
- Debugging Help: When I ran into strange output mismatches or edge-case behavior, I used AI to search for possible causes and suggest debugging steps. I did tests and compared results to the project specifications to validate proposals rather than just copying answers.